

TrES-3b

R-2813

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_120825 · created 2026-09-05T01:34:09+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456164.7843 +/- 0.001 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.012 +/- 0.055
Transit depth (Rp/Rs)^2	0.015 +/- 0.14 [%]
Orbital Inclination (inc)	81.21 +/- 2.81
Airmass coefficient 1 (a1)	0.991 +/- 0.036
Airmass coefficient 2 (a2)	0.009 +/- 0.022
Scatter in the residuals of the lightcurve fit is	3.65 %
Best Comparison Star	#2 - [135, 117]
Optimal Aperture	2.84
Optimal Annulus	9.91
Transit Duration (day)	0.029 +/- 0.026

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.012 ± 0.055 vs 0.16309 (deviation 2.75σ)
Duration fit vs published	0.029 d vs 0.0567 d (deviation 1.06σ)
Mid-time O–C	0.7 min
Depth SNR	0.2
Residual scatter	3.65 %

Light curve

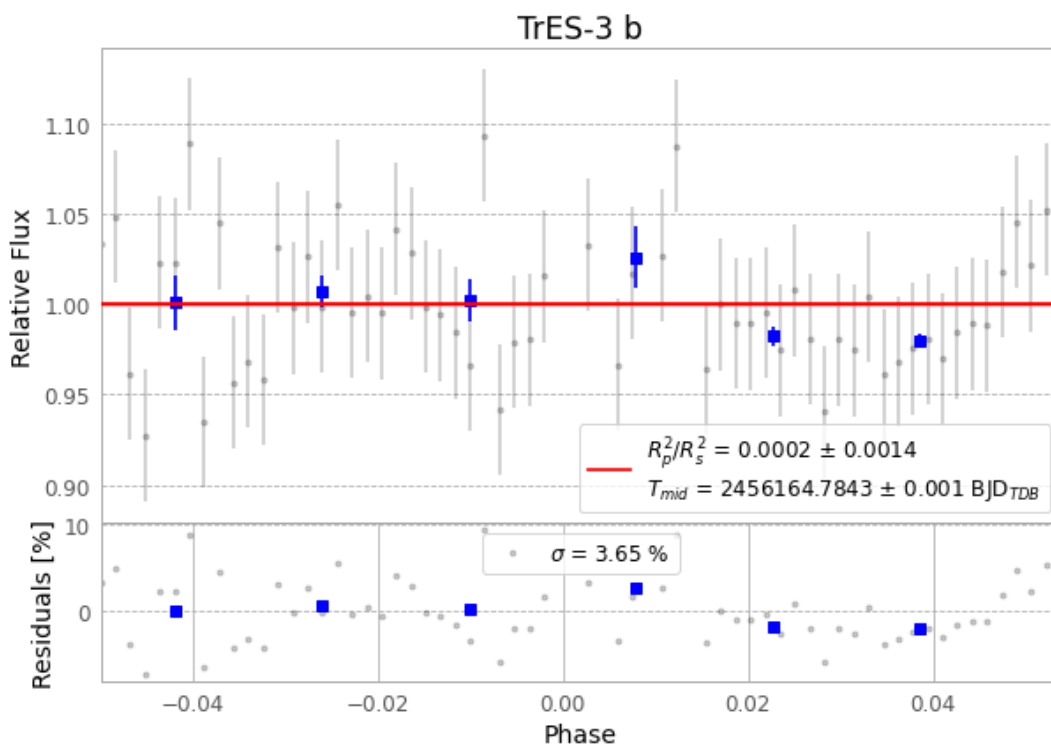


Figure 1 — FinalLightCurve_TrES-3 b_2012-08-24.png (SHA-256 2b521bc5de8967a1...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [207, 275], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 129b8c932c66428d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

66 FITS frames (43 MB), TRES-3120825051212.FITS → TRES-3120825082712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-120825.log (4513c06e30a2...), FinalLightCurve_TrES-3 b_2012-08-24.png (2b521bc5de89...), FinalLightCurve_TrES-3 b_2012-08-24.csv (d40ceeed74a3...)

Compute resources

Wall time 182.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-05 01:34Z.